

MSTAR PRACTICE WORKSHEET**Lesson 7-7: Represent Polygons on the Coordinate Plane**

Grade 6 Mathematics · Standard 6.NOS.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: D**Vertices $(-4.5, 3)$ and $(5, 3)$ share $y = 3$. What is the distance between them?

- ☐ A 0.5 units
- ☐ B 1.5 units
- ☐ C 8 units
- ☒ D 9.5 units

They differ only in x: $|5 - (-4.5)| = 5 + 4.5 = 9.5$ units.**Question 2** **SELECTED RESPONSE — CORRECT: C**Vertices $(5, 3)$ and $(5, -3.75)$ share $x = 5$. What is the distance between them?

- ☐ A 0.75 units
- ☐ B 5 units
- ☒ C 6.75 units
- ☐ D 8.75 units

They differ only in y: $|3 - (-3.75)| = 3 + 3.75 = 6.75$ units.

Question 3 **SELECTED RESPONSE — CORRECT: C**

What is the perimeter of the rectangle with side lengths 9.5 and 6.75 units?

- ☐ A 16.25 units
- ☐ B 26 units
- ☒ C 32.5 units
- ☐ D 64.125 units

Perimeter = $2(9.5) + 2(6.75) = 19 + 13.5 = 32.5$ units.

Question 4 **SELECTED RESPONSE — CORRECT: D**

Why are two of the terms multiplied by 2 in the perimeter calculation?

- ☐ A Because perimeter always doubles the area
- ☐ B Because there are two coordinates in each ordered pair
- ☐ C It is a rounding rule
- ☒ D A rectangle's opposite sides are equal, so each length appears on two sides

The two 9.5 sides and the two 6.75 sides each come in pairs — that is what the 2 counts.

Question 5 **SELECTED RESPONSE — CORRECT: D**

To draw a polygon from a list of vertices, you:

- ☐ A Plot only the first vertex
- ☐ B Connect the points in any random order
- ☐ C Add all the coordinates together
- ☒ D Plot each ordered pair, then connect them in order with segments

Graph the points on the plane and connect them in order — the closed figure that appears is the polygon.

Question 6

SELECTED RESPONSE — CORRECT: D

The park's marked vertices are 3 units apart and each unit is 100 feet. What is the park's width?

- ☐ A 3 feet
- ☐ B 30 feet
- ☐ C 103 feet
- ☒ D 300 feet

3 units $\times$ 100 feet per unit = 300 feet.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

A rectangle has vertices at (1, 1), (7, 1), (7, 5), and (1, 5). What is its perimeter?

- ☐ A 10 units
- ☐ B 12 units
- ☒ C 20 units
- ☐ D 24 units

The horizontal sides are $|7 - 1| = 6$ units and the vertical sides are $|5 - 1| = 4$ units. Perimeter = $2(6) + 2(4) = 20$ units.

- **A:** That adds one of each side. Each length appears twice around a rectangle.
- **B:** That doubles the horizontal side only. The two vertical sides of 4 units each still count.
- **D:** That is the AREA, 6 times 4, in square units. Perimeter adds the four side lengths.

Part B — correct answer: B

How can the side lengths be found from the coordinates alone?

- ☐ A Count every grid square inside the rectangle.
- ☒ B Each side is horizontal or vertical, so its length is the difference of the coordinates that change.
- ☐ C Add all eight coordinate values together.
- ☐ D Multiply the x-coordinates by the y-coordinates.

Every side of this rectangle runs along a grid line, so one coordinate stays fixed and the length is the absolute difference of the other.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about polygons on the coordinate plane:

- ☒ **A** Vertices are connected in the order given to form the polygon.
- ☒ **B** A horizontal side's length is the absolute difference of the x-coordinates.
- ☒ **C** A vertical side's length is the absolute difference of the y-coordinates.
- ☐ **D** The length of a slanted side can be found by subtracting coordinates.
- ☒ **E** The area of a rectangle on the grid is the product of its side lengths.

A, B, C, and E are true. D is false — a slanted side changes in BOTH directions, so subtracting one pair of coordinates does not give its length.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

The park design used width from coordinates, then length from the area, then the scale to place the missing vertices. Design your own version: pick an area and a scale, mark two vertices, and show where the other two go.

Model answer: My park has area 60,000 sq ft with each unit = 100 ft. I mark vertices at (1, 5) and (3, 5) — 2 units apart, so width 200 ft. Length = $60,000 \div 200 = 300$ ft = 3 units, so the other vertices go at (1, 2) and (3, 2), 3 units below.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.